

Advanced Machining Processes in Modern Manufacturing: A Critical Review

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Abstract

Advanced machining processes (AMPs) have revolutionized modern manufacturing by enabling precise, efficient, and sustainable material removal in hard-to-machine materials and complex geometries. This paper presents a comprehensive review of AMPs, systematically classifying them into mechanical, thermoelectric, and electrochemical/chemical categories based on their energy utilization and material removal mechanisms. The study critically examines key processes including abrasive jet machining (AJM), electric discharge machining (EDM), laser beam machining (LBM), and electrochemical machining (ECM), analyzing their principles, capabilities, and industrial applications. Significant attention is given to recent advancements such as hybrid processes, micro/nano-machining techniques. The review highlights crucial process parameters affecting performance metrics like material removal rate, surface finish, and dimensional accuracy, while discussing optimization strategies through experimental and computational methods.

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1. INTRODUCTION

Electrical Discharge Machining (EDM) is a controlled metal-removal process that is used to remove metal by means of electric spark erosion [1]. In this process an electric spark is used as the cutting tool to cut (erode) the work piece to produce the finished part to the desired shape. The metal-removal process is performed by applying a pulsating (ON/OFF) electrical charge of high-frequency current through the electrode to the work piece. This removes (erodes) very tiny pieces of metal from the work piece at a controlled rate [2]. Electric Discharge Machining is also called as Spark Erosion Machining, Die Sinking, Spark-over-initiated Discharge Machining, Electro-erosion Machining, Wire Erosion or simply Spark Machining.

EDM is the process of electrically removing material from any conductive work piece. This is achieved by applying high-frequency pulsed, AC or DC current to the work piece through an electrode or wire, which melts and vaporizes the work piece material. Positioned very precisely near the work piece, the electrode never touches the work piece but discharges its potential current through an insulating dielectric fluid across a very small spark gap [3]. The spark is reported to be in the range of 8000 to 12000°C (14432 to 21632°F), and it vaporizes and melts the work piece material. This process is used when the work piece material is too hard, or the shape or location of the detail cannot easily be conventionally machined. This makes many formerly difficult projects more practical, and many times it can be the only feasible way to machine a part or material [4-9].

EDM was first implemented over 40 years ago, and used primarily to remove broken taps and drills from expensive parts. These were quite crude in construction with hand fed electrodes. During World War II, two Russian scientists, B. R. and N. I. Lazarenko, adapted the first servo – system to an EDM machine [10].

The WEDM is rather a newer concept in Electro discharge machining, where a very thin brass or copper wire is used instead of the electrode [11].

This wire, which varies from 0.02 mm to 0.03 mm in diameter, is fed continuously into the machining area under the water beam. In this process only through machining is possible where a small hole is drilled in the work piece and the same mounted on the work table. The feed wire is passed through this hole and held tight between the feeding and pickup units so that it runs in a straight line in the machining area by following the programmed contour. Thus the machining is carried out rest of the process is almost similar to EDM [12].

Wire electrical discharge machining (WEDM) is a widely accepted non-traditional material removal process used to manufacture components with intricate shapes and profiles. It is considered as a unique adaptation of the conventional EDM process, which uses an electrode to initialize the sparking process. However, WEDM utilizes a continuously travelling wire electrode made of thin copper, brass or tungsten of diameter 0.05–0.3 mm, which is capable of achieving very small corner radii. The wire is kept in tension using a mechanical tensioning device reducing the tendency of producing inaccurate parts. During the WEDM process, the material is eroded

ahead of the wire and there is no direct contact between the work piece and the wire, eliminating the mechanical stresses during machining [13].

Wire electrical Discharge Machining (WEDM) is defined as the non-traditional process of material removal of electrically conductive materials to produce the part with intricate shapes and profiles. This process is done by using a series of spark erosion process. These spark erosion is produced between the work piece and a wire electrode which is separated by a dielectric fluid. The sparks produced heating and melting work piece surface to form debris then it is flushed away by dielectric pressure. During the cutting process there is no direct contact between the work piece and the wire electrode [14].

To get the perfect result of the machining process by using the WEDM we need to find the correct parameter setting. Until now, there is no perfect parameter setting for any type of materials. So, it is important to find the best parameter setting before start the machining process in order to achieve the maximum result in its material removal rate (MRR) and surface roughness (Ra).

Advantages

- It has ability to create complex and intricate parts with a high degree of accuracy.
- This process is able to machine hard materials that other machining processes would have difficulties.
- Another advantage of WEDM is its ability to machine parts on an extremely small scale.
- While using this process, the workpiece is not deformed from impact because there is no direct contact between the electrode and the material.
- The work piece is burr-free after completion and saved from heat damage because very little is actually generated during the procedure that would harm the material.

2. LITERATURE SURVEY

Table 1: Different studies o EDM

Sr.N o.	Title	Author's Name	Year	Material	Input Parameter	Output Parameter	Most Significant
1	Optimization of Wire EDM Parameters of EN 24 Steel by Taguchi Method	Sheril Cyriac	2015	EN 24	Pulse time, Pulse of Speed	Surface Roughness	Current
2	Wire electro-discharge machining of titanium alloy	Farnaz Nourbakhs h	2013	Titanium Alloy, zinc-	Pulse time, Pulse	MRR	T-on

				coated brass wir	of time, voltage		
3	Optimization of WEDM parameters using Taguchi method for higher material removal rate on EN31 steel	Dipti Ranjan Patra	2015	EN31 steel	Gap Voltage,Wir e feed, Pulse on time, Pulse of time,	Material removal rate	Pulse on time
4	Optimization of process parameters using taguchi's design of experiment in wire cut electrical discharge machining	Jayaram M.	2018	SS 304	Pulse on time, Pulse of time,Peak current, wire tension,Wwi re feed, Fluid Pressure	Surface Roughness	Peak Current
5	Optimization of Process Parameters in Electrical Discharge Machining of EN 8 Steel by Using Taguchi Technique	L. Feroz Ali	2019	EN 8 Alloy	Current, Pulse on time.Pulse off time	Material removal rate, Electroded Wear Rate,SR	Current
6	Optimization of Wire Electro Discharge Machining of HCHCr Material using Taguchi Methodology	Kranti Kumar Bhosale1,	2017	High Carbon High Chromium (HCHCr	Pulse on time.Pulse off time,feed rate,sensitivitiy	Surface Roughness , Material removal rate	T-off
7	Optimization of machining parameters in WEDM of AISI D3 Steel using Taguchi Technique	Bra jesh Kumar Lodhia	2014	AISI D3 Steel	Pulse on time, Pulse of time current,wire feed	Surface Roughness	Current
8	Optimisation and Influence of Process Parameters for Machining with Wire Cut EDM Using Taguchi's Technique	Shrey Kapoor	2018	HSS	Pulse on time, Pulse of time current,Serv o voltage	Material removal rate	T-on
9	Optimization of machining parameters of wire cut edm process using taguchi method	Pawan Kumar	2015	Steel H13	Pulse on time, Pulse of time current,Wire Feed, Wire Tension	Material removal rate, Surface Roughness	Current

Conclusion from the Literature Survey

From an extensive review of research papers on Wire Electrical Discharge Machining (WEDM), it is evident that various types of wires with distinct operating characteristics have been employed

to enhance machining performance, including Material Removal Rate (MRR), kerf width, surface finish, and striation effects. Several modifications and optimizations in process parameters have been found beneficial in improving machining efficiency and precision. Key observations from the literature include:

Experimental studies have demonstrated that predicted values for MRR and surface roughness closely align with actual experimental results, indicating the reliability of theoretical models in WEDM. The primary parameters influencing MRR, kerf width, and surface roughness include cutting current, pulse-off time, wire speed, pulse-on time, feed rate, sensitivity, fluid pressure, servo voltage, dielectric flow rate, pulse duration, and pulse frequency. Among these, cutting current, pulse-on time, wire speed, pulse-off time, and servo voltage play the most significant roles in determining surface quality and machining speed.

Additionally, the literature highlights the impact of electrode materials on machining performance. While copper, brass, and graphite electrodes are commonly used, limited research has been conducted on molybdenum wire electrodes, suggesting a potential area for further investigation. Although existing models may not be universally applicable across all machining conditions, the methodologies employed provide a strong foundation for deeper process analysis.

Most studies in this domain have focused on optimizing a single quality characteristic at a time, such as MRR or surface finish, rather than adopting a multi-objective optimization approach. This indicates a need for future research to explore integrated optimization techniques that simultaneously improve multiple machining responses. Overall, the findings underscore the importance of parameter selection and electrode material in WEDM, paving the way for advancements in precision machining for complex and high-performance applications.

3. Classification of Advanced Machining Processes

Advanced machining processes (AMPs) represent a specialized group of manufacturing techniques designed to overcome the limitations of conventional machining. These processes are particularly valuable for working with hard, brittle, or complex materials that traditional methods struggle to process effectively. Unlike conventional techniques that rely primarily on mechanical cutting forces, AMPs utilize various energy forms including mechanical, thermal, electrical, and chemical energy to achieve material removal. The classification into mechanical, thermoelectric, and electrochemical/chemical categories reflects the fundamental energy source and mechanism

behind each process. This systematic categorization helps engineers and manufacturers select the most appropriate method based on material properties, desired precision, and economic factors.

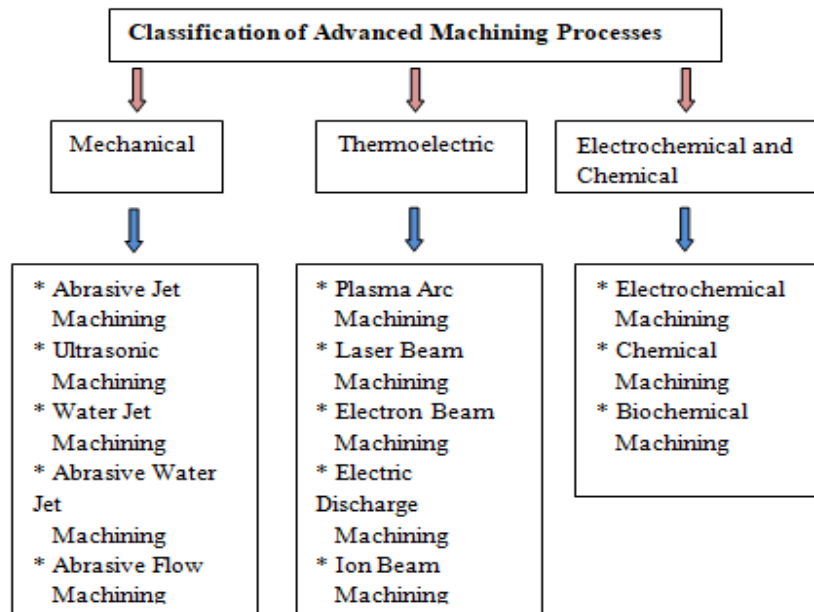


Figure 1 Classification of Advanced Machining Processes

The diverse landscape of advanced machining processes offers tailored solutions for modern manufacturing challenges across industries. From mechanical methods preserving material properties to thermoelectric processes handling extreme hardness, and chemical techniques enabling atomic-level precision, each category addresses specific needs in contemporary production. The ongoing evolution of these technologies focuses on hybrid approaches combining multiple energy forms, enhanced precision through digital controls, and sustainable alternatives reducing environmental impact.

4. Key Process Parameters in EDM

Table 2: The performance of EDM several parameters

Parameter	Effect on Machining
Discharge Current	Higher current increases material removal rate (MRR) but reduces surface finish.
Pulse Duration (Ton/Toff)	Longer pulse-on time (Ton) increases MRR but causes higher tool wear.
Voltage	Affects spark gap and machining stability.
Dielectric Fluid	Influences debris removal and cooling efficiency.
Electrode Material	Graphite (high MRR), Copper (fine finish), Tungsten (wear resistance).

5. Conclusion on Advanced Machining Processes

This comprehensive review has systematically examined the classification, mechanisms, applications, and recent advancements in advanced machining processes (AMPs). As modern manufacturing faces increasing demands for precision, complex geometries, and difficult-to-machine materials, AMPs have emerged as indispensable solutions that overcome the limitations of conventional machining techniques.

Key findings from this review highlight several critical aspects:

- i. Process selection depends fundamentally on material properties, required precision, and economic considerations
- ii. Hybrid approaches combining multiple energy forms are demonstrating superior performance
- iii. Precision capabilities continue to advance, with some processes now operating at nanometer scales
- iv. Environmental sustainability is becoming a driving factor in process development

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